Appendix C: IEEE 830 Template

Software Requirements Specification
for
ESP32 Room Climate Station

Version 0.1 (Draft)

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April 27, 2026

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Revision History

Name	Date	Reason for Changes	Version
Ihor Huz	April 27, 2026	Initial draft for Lab 1: Requirement Analysis	0.1

1. Introduction

1.1. Purpose

«Identify the product whose software requirements and design are specified in this document, including the revision or release number. Describe the scope of the product that is covered by this specification.»

1.2. Document Conventions

«Describe any standards or typographical conventions that were followed when writing this specification, such as fonts or highlighting that have special significance. For example, state whether priorities for higher-level requirements are assumed to be inherited by detailed requirements, or whether every requirement statement is to have its own priority.»

1.3. Intended Audience and Reading Suggestions

«Describe the different types of reader that the document is intended for, such as developers, project managers, marketing staff, users, testers, and documentation writers. Describe what the rest of this document contains and how it is organized. Suggest a sequence for reading the document, beginning with the overview sections and proceeding through the sections that are most pertinent to each reader type.»

1.4. Product Scope

The ESP32 Room Climate Station addresses this limitation by functioning as a centralized, persistently logging IoT environment monitor. It bridges the gap between local hardware and remote software by reading ambient conditions through a high-accuracy SHTC3 sensor and making that data available through three distinct interfaces: a local LCD, a web-based dashboard, and a Telegram bot. Furthermore, the system solves the common IoT problem of volatile memory loss by utilizing SPIFFS to store a 24-hour rolling buffer of environmental data. This ensures that in the event of a power failure, the user does not lose their historical trend data. The integration of mDNS and Over-The-Air (OTA) updates ensures the device remains a low-maintenance, headless appliance once deployed.

1.5. References

«List any other documents or Web addresses to which this document refers or was based on. These may include user interface style guides, contracts, standards, system requirements specifications, use case documents, or a vision and scope document. Provide enough information so that the reader could access a copy of each reference, including title, author, version number, date, and source or location. Include the assignment document in the references.»

2. Overall Description

2.1. Product Perspective

Maintaining optimal indoor climate conditions (temperature and humidity) is critical for personal health, equipment longevity, and general comfort. Traditional consumer thermometers provide only real-time snapshots of the environment, lacking historical context and remote accessibility. When users leave the room or building, they lose visibility into the environmental trends.

2.2. Product Features

«Summarize the major features the product must perform or must let the user perform. Details will be provided in Section 3, so only a bullet list is needed here. Organize the features to make them understandable to any reader of the specification. A picture of the major groups of related requirements and how they relate, such as a top level data flow diagram or object class diagram, is often effective.»

- «Thing 1»
- «More things as required»

2.3. User Classes and Characteristics

«Identify the various user classes (intended target for the product) that you anticipate will use this product. User classes may be differentiated based on frequency of use, subset of product functions used, technical expertise, security or privilege levels, educational level, or experience. Describe the pertinent characteristics of each user class. Certain requirements may pertain only to certain user classes. Distinguish the most important user classes for this product from those who are less important to satisfy.»

2.4. Operating Environment

«Describe the environment in which the software will operate, including the hardware platform, operating system and versions, and any other software components or applications with which it must peacefully coexist.»

2.5. Design and Implementation Constraints

«Describe any items or issues that will limit the options available to the developers. These might include: corporate or regulatory policies; hardware limitations (timing requirements, memory requirements); interfaces to other applications; specific technologies, tools, and databases to be used; parallel operations; language requirements; communications protocols; security considerations; design conventions or programming standards (for example, if the customer's organization will be responsible for maintaining the delivered software).»

2.6. User Documentation

«List the user documentation components (such as user manuals, on-line help, and tutorials) that will be delivered along with the software. Identify any known user documentation delivery formats or standards.»

2.7. Assumptions and Dependencies

«List any assumed factors (as opposed to known facts) that could affect the requirements stated in the specification. These could include third-party or commercial components that you plan to use, issues around the development or operating environment, or constraints. The project could be affected if these assumptions are incorrect, are not shared, or change. Also identify any dependencies the project has on external factors, such as software components that you intend to reuse from another project, unless they are already documented elsewhere (for example, in the vision and scope document or the project plan).»

3. System Features

«The purpose here is to give a brief introduction to the system features, so that §4 has something to reference. Then, the full explanation of the system features is given in §5, likely referencing §4. The numbering in §3 and §5 should match.»

3.1. «Feature Name»

«Brief description of the feature, generally 1-3 sentences»

3.«N». «Additional feature blocks as needed»

4. External Interface Requirements

4.1. User Interfaces Overview

«Describe the high level functionality of the system from the user's perspective. Explain how the user should be able to use your system to complete all the expected features and the feedback information that will be displayed for the user (for each screen the user will see). For example, specify whether it is GUI or text based interface and how at high level it would work. Discuss at high level how user will interact with the game such as clicking buttons, typing in some selection character, etc.»

«Describe the minimum requirements for the interface. This may include some sample screen images whether for text or GUI approach, any GUI standards or product family style guides that are to be followed, screen layout constraints, for GUI standard buttons and functions (e.g., help) that must appear on every screen, keyboard shortcuts, error message display standards, and so on»

4.2. Hardware Interfaces

«Describe the logical and physical characteristics of each interface between the software product and the hardware components of the system. This may include the supported device types, the nature of the data and control interactions between the software and the hardware, and communication protocols to be used.»

4.3. Software Interfaces

«Describe the connections between this product and other specific software components (name and version), including databases, interpreters, operating systems, tools, libraries, and integrated commercial components. Identify the data items or messages coming into the system and going out and describe the purpose of each. Describe the services needed and the nature of communications. Identify data that will be shared across software components. If the data sharing mechanism must be implemented in a specific way (for example, use of a global data area in a multitasking operating system), specify this as an implementation constraint.»

5. System Features/Modules

«This template illustrates organizing the functional requirements for the product by system features, the major services provided by the product.»

5.1. «System Feature Name»

5.1.1 Description and Priority

«Provide a short description of the feature. Give its priority, too.»

5.1.2 Stimulus/Response Sequences

«"Stimulus" or "Condition"»: «Describe the stimulus or condition»

Response: «Describe the response»

«Additional Stimulus/Response pairs as needed»

5.1.3 Functional Requirements

«Itemize the detailed functional requirements associated with this feature. These are the software capabilities that must be present in order for the user to carry out the services provided by the feature. Include how the product should respond to anticipated error conditions or invalid inputs. Requirements should be concise, complete, unambiguous, verifiable, and necessary. Each requirement should be uniquely identified with a sequence number or a meaningful tag of some kind.»

Feature 1: Local Climate Monitoring and Display

REQ-1.1: The system shall measure ambient temperature and humidity using the connected SHTC3 sensor.

REQ-1.2: The system shall display the current temperature and humidity readings locally on the 16x2 I2C LCD.

REQ-1.3: The system shall allow the user to manually toggle the LCD backlight state (on/off) using a physical hardware button connected to GPIO 4.

Feature 2: Telegram Bot Integration

REQ-2.1: The system shall automatically transmit a formatted ambient climate report to the configured Telegram Bot every 15 minutes.

REQ-2.2: The system shall parse and execute the following remote commands received via Telegram: `/light_on` (enables LCD), `/light_off` (disables LCD), `/status` (returns current readings), and `/restart` (initiates a soft reboot).

REQ-2.3: The system shall clear the Telegram message queue immediately upon booting to prevent infinite restart loops triggered by stale commands.

Feature 3: Web Dashboard and Device Management

REQ-3.1: The system shall serve a local web dashboard accessible via `http://climate.local` that displays a Highcharts-powered visualization of both live trends and 24-hour historical data.

6. Nonfunctional Requirements

6.1. Performance

NF-1.1: The system shall utilize non-blocking timers for network operations to ensure that the physical hardware toggle (GPIO 4) and local LCD updates remain responsive, acting with less than a 500-millisecond delay.

6.2. Reliability

NF-2.1: The system shall persist a rolling 24-hour history of sensor data to internal SPIFFS flash memory to ensure data is preserved and reloaded across unexpected device restarts.

6.3. Maintainability

NF-3.1: The system shall support Over-The-Air (OTA) firmware updates via a local web browser interface, allowing deployment of new code without a physical USB connection.